

BE IT 2nd YEAR 2nd SEM-2026- OBJECT ORIENTED SYSTEMS

Time: 1 hour

CT2- SET-B

Full marks: 30

1. a) Demonstrate how deadlock situation occurs in a multi-threaded environment. Show how this can be eliminated.

b) Illustrate the functionalities of the following functions related to thread. State all the overloaded versions of each of the function (if exist): join(), interrupt(), sleep(), wait(), notify()

[(5+2)+10=17]

2. a) Define a generic class having 3 overloaded versions of a generic method. One of them should be static. Now show with a suitable program how to invoke these methods from main(). Also show how a parameterized constructor of this class can be invoked from the body of another constructor of that same class.

b) Distinguish between each of the following pairs with suitable code snippets.

i) FileInputStream and BufferedInputStream ii) nextInt() and hasNextInt()

[(5+2)+(3+3)=13]

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Time: 1 hour

CT2-SET-A

Full marks: 30

1. a) What is the drawback of creating a thread extending the Thread class? State how this drawback can be overcome. Show how the execution of a thread can be started?

b) What is the purpose of declaring a block as synchronized? Clearly distinguish synchronized block from synchronized method and indicate which one is better to use in which scenario/s.

[(2+3+2)+(2+2+2)=13]

2. a) Point out the drawback of Scanner class with necessary code snippets. Provide 2 suitable ways to overcome the drawback.

b) Show how you can define a static generic function in a class taking any type of parameter as argument. Also show how the method can be invoked.

c) Define a generic class with a member method Sorting(A) that take arrays A of different datatypes (int, float, char) as input and sorts the array using Selection sorting algorithm.

[(3+4)+3+7=17]